

1.Product Purchase (Online Shopping)

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Product Available	Yes	No	Yes	Yes
User Logged In	Yes	Yes	No	Yes
Payment Successful	Yes	Yes	Yes	No
Action	Order Placed	Product Out of Stock	Login Required	Payment Failed

Test Cases-

Test Case ID	Product Available	User Logged In	Payment Status	Expected Result
TC13	Yes	Yes	Success	Order is placed successfully and confirmation message is displayed.
TC14	No	Yes	Success	System displays 'Product Out of Stock' and order is not placed.
TC15	Yes	No	Success	System redirects the user to the login page before checkout.
TC16	Yes	Yes	Failed	Payment fails and the order is not confirmed.

Explanation

This decision table verifies the order placement functionality of an online shopping application. The order is completed only when all three conditions are satisfied:

- The selected product is available in stock.
- The customer is logged into the application.
- The payment transaction is successful.

If any one of these two conditions is not satisfied, the system displays the appropriate error message and prevents the order from being placed.

2.Password Reset

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Registered Email	Yes	Yes	No	Yes
OTP Correct	Yes	No	Yes	Yes
New Password Valid	Yes	Yes	Yes	No
Action	Password Reset Successfully	Invalid OTP	Email Not Found	Weak Password

Test Cases

Test Case ID	Email	OTP	New Password	Expected Result
TC17	Registered	Correct	Strong	Password is reset successfully and the user can log in with the new password.
TC18	Registered	Incorrect	Strong	System displays 'Invalid OTP' and does not reset the password.
TC19	Unregistered	Correct	Strong	System displays 'Email Not Found.'
TC20	Registered	Correct	Weak	System displays a password policy message and rejects the new password.

Explanation

This decision table validates the password recovery feature. Password reset is allowed only when:

- The email address exists in the system.
- The OTP entered by the user matches the generated OTP.

- The new password satisfies the password policy(minimumlength,uppercase,lowercase, number, and special character).

If any condition fails,the system displays the appropriate error message and prevents the password from being reset.

3.ATM Cash Withdrawal

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Valid ATM Card	Yes	Yes	No	Yes
Correct PIN	Yes	No	Yes	Yes
Sufficient Balance	Yes	Yes	Yes	No
Action	Cash Withdrawn	Invalid PIN	Invalid Card	Insufficient Balance

Test Cases

Test Case ID	ATM Card	PIN	Balance	Expected Result
TC21	Valid	Correct	Sufficient	Cash is withdrawn successfully and balance is updated.
TC22	Valid	Incorrect	Sufficient	System displays 'Invalid PIN'.
TC23	Invalid	Correct	Sufficient	System displays 'Invalid ATM Card'.
TC24	Valid	Correct	Insufficient	System displays 'Insufficient Balance'.

Explanation

The ATM withdrawal feature checks three conditions before processing a transaction:

- The ATM card must be valid.
- The PIN entered must match the registered PIN.
- The account should have enough balance.

If any condition fails, the transaction is rejected with an appropriate error message.

4.Railway Ticket Booking

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Seats Available	Yes	No	Yes	Yes
Payment Successful	Yes	Yes	No	Yes
Passenger Details Valid	Yes	Yes	Yes	No
Action	Ticket Booked	No Seats Available	Payment Failed	Invalid Passenger Details

Test Cases

Test Case ID	Seats	Payment	Passenger Details	Expected Result
TC25	Available	Success	Valid	Ticket is booked successfully and PNR is generated.
TC26	Not Available	Success	Valid	System displays 'No Seats Available'.
TC27	Available	Failed	Valid	Ticket booking fails due to unsuccessful payment.
TC28	Available	Success	Invalid	System asks the user to correct passenger details.

Explanation

This decision table verifies the railway booking process. A ticket is confirmed only when:

- Seats are available.
- Payment is completed successfully.
- Passenger details are valid.

If any one of these conditions is not satisfied, the booking process is stopped.

5. Online Food Order

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Restaurant Open	Yes	No	Yes	Yes
Food Available	Yes	Yes	No	Yes
Payment Successful	Yes	Yes	Yes	No
Action	Order Confirmed	Restaurant Closed	Item Unavailable	Payment Failed

Test Cases

Test Case ID	Restaurant	Food Item	Payment	Expected Result
TC29	Open	Available	Success	Order is confirmed and delivery time is displayed.
TC30	Closed	Available	Success	System displays 'Restaurant is Closed'.
TC31	Open	Not Available	Success	System displays 'Selected Item is Unavailable'.
TC32	Open	Available	Failed	Payment fails and the order is not placed.

Explanation

The food ordering system validates three conditions:

- The restaurant should be open.
- The selected food item must be available.
- Payment should be completed successfully.

Only when all conditions are satisfied is the order confirmed; otherwise, the appropriate error message is shown.